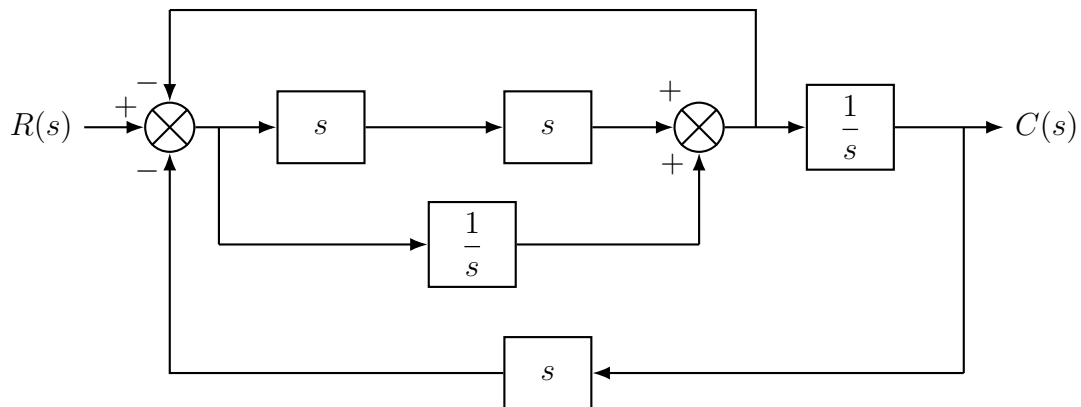
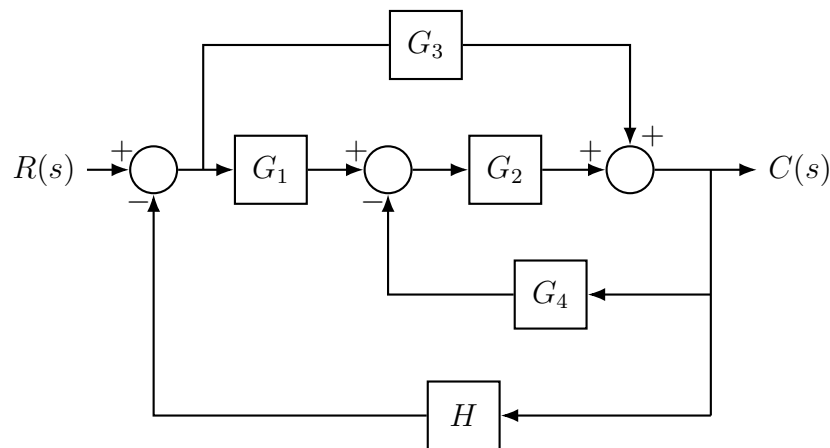


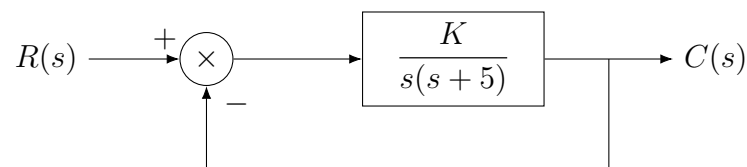
Q1. Find the equivalent transfer function, $T(s) = C(s)/R(s)$, for the system shown in Figure given below.



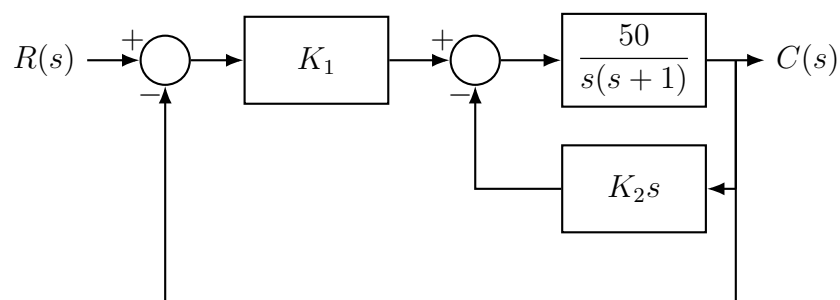
Q2. Reduce the system shown in Figure to a single transfer function, $T(s) = C(s)/R(s)$.



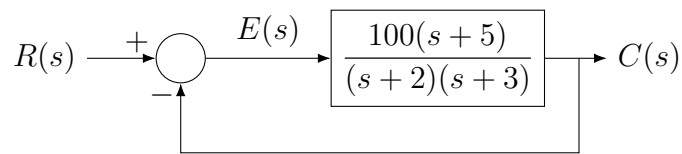
Q3. Design the value of gain K , for the feedback control system of Figure so that the system will respond with a 10% overshoot.



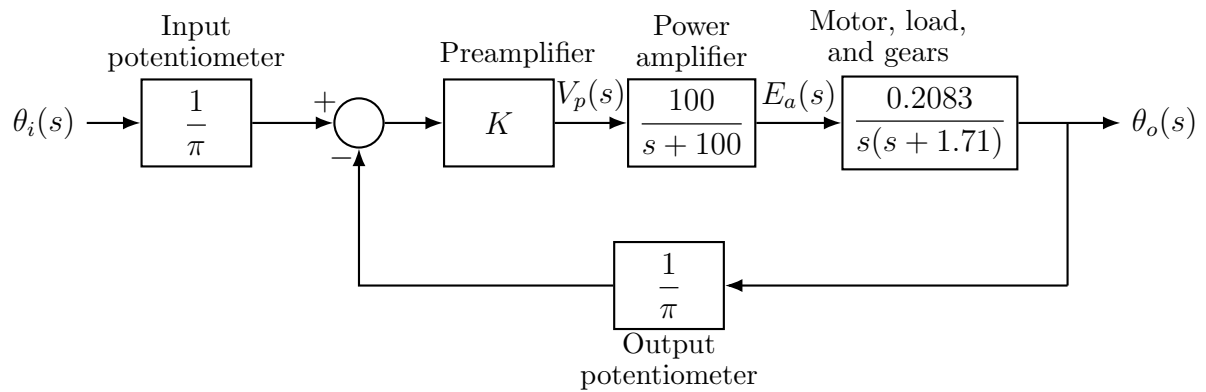
Q4. For the system, find the values of K_1 and K_2 to yield a peak time of 1 second and a settling time of 2 seconds for the closed-loop system's step response.



Q5. Represent the feedback control system shown in Figure in state space.



Q6. The antenna azimuth position control system is shown in Figure given below.



Convert the system into unity feedback system and determine the closed loop transfer function and state-space model.